

Hydraulics

(Coach Scheduled Event - 45 Minutes)



Description: Each team will build a device that uses a hydraulic system to accomplish the task of lifting a 100g weighted object as high as possible while holding the weight for as long as possible. The team will also take a written test on basic hydraulic concepts.

Participants per assigned Team Number: 2

- If your school has 1 team you will send 2 students
- If your school has 2 teams you will send 2 students per team number; team numbers may not intermix
- If your school has 3 teams you will send 2 students per team number; team numbers may not intermix

Safety Equipment - Teams Must Bring:

- **Safety glasses labeled ANSI Z87+ (impact rated)**
- **All competitors must wear their eye protection during the entire event.**
- **If a team does not have the required eye protection, they will be given the opportunity to obtain it, time allowing, but will not receive extra time.**
- **If a team is unable to obtain eye protection, the team will not compete and will receive a no-show score.**

Impact Safety Glasses Required



Eyeglasses are not safety rated!

Team members must also bring:

- Pencils
- Scissors
- Teams may bring a straight edge measuring device

Materials Provided at the Event:

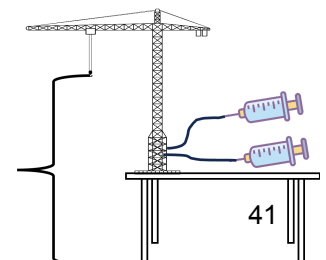
- 1 meter (39 in) of masking tape
- 1 meter (39 in) of 2 mm (0.08 in) string/kitchen twine + enough extra to span the distance between the floor and the table top their device sits upon
- 10 paper-fastener brass brads
- 2-12 ml (0.4 oz) syringes
- 50 cm (19.5 in) of PVC Tubing 3.175mm ($\frac{1}{8}$ in) inside diameter X 6.35mm ($\frac{1}{4}$ in) outside diameter Flexible Clear Vinyl Hose
- 20 Craft Sticks with Holes 15 cm (6 in)
- 1 Spool $\approx$ 2.54 cm (1 in) x 1.9 cm (0.75 in) with a center hole $\approx$ 6 mm ($\frac{1}{4}$ in)
- Water
- 100g mass
- 1 - large paperclip

THE COMPETITION

Written Assessment: The team members will take a test on the principles of hydraulics. The test will occur concurrently with the build.

Construction Phase (30 min):

- Each team will have 30 minutes to construct their device.



- The teams do not have to use all of their materials but at a minimum use both syringes.
- Only those materials provided may be used to build the device.
- The device must be freestanding and may not be attached to a tabletop, floor, ceiling or other support.
- The team will be responsible for filling their syringes and tubing with the supplied water.

Device Testing:

- The 100 gram weight will be placed on the floor, while the team's device will stay on their table.
- The team will tie the end of the string that dangles to the floor to a loop on the weight.
- When ready, with the weight resting on the floor, the team will give a countdown so the Event Supervisor may begin timing. The Event Supervisor will begin the time once the weight lifts off the floor.
- The team will lift the weight with their device and hold it off the floor as long as possible (up to a max of 30 seconds).
- The Event Supervisor will measure the maximum height the weight was lifted. The height will be measured from the floor to the bottom of the weight.
- The Event Supervisor will time how long the weight is lifted off the floor. The maximum time recorded will be 30 seconds.

SCORING Final Score (FS) = Assessment Score + Height Score + Time Score, max = 100 pts

- Assessment score = $(TAS \div CAS) \times 20$; rounded to the 100th place value; maximum 20 points
 - Team Assessment Score (TAS) = team's score on the assessment
 - Competition Assessment Score (CAS) = the highest assessment score for all teams
- Height Score = $(THS \div CHS) \times 50$; rounded to the 100th place value; maximum 50 points
 - Team Height Score (THS) = the highest distance of lift for all teams
 - Competition Height Score (CHS) = the highest distance of lift over all teams
- Time Score = $(TTS \div CTS) \times 30$; rounded to the 100th place value; maximum 30 points
 - Team Time Score (TTS) = team time for weight off the floor (max 30 seconds)
 - Competition Time Score (CTS) = longest time of all teams with weight off the floor (max 30 sec)

Example

- Competition Assessment Score (CAS) = 15
- The team scores 9 (TAS) on the assessment.
 - Assessment Score = $(9 \div 15) \times 20 = 12$ points
- Competition Height Score (CHS) = 35 cm
- The team height (THS) is 25 cm
 - Height Score = $(25 \div 35) \times 50 = 35.71$ points
- Competition Time Score (CTS) = 27 seconds
- The team's time (TTS) is 1.45 seconds
 - Time Score = $(1.45 \div 27) \times 30 = 1.61$ points
- Final Score (FS) = $12 + 35.71 + 1.61 = 49.32$ points

Tiebreakers:

1. Best written test score
2. Selected questions on the written test
3. Best Competition Height

Resources:

- [Young Engineers: Easy Hydraulic or Pneumatic Machine - Engineering Projects for Kids](#)
- [Modern Marvels: Hydraulic Force Transforms Society \(S11, E17\) | Full Episode | History](#)
- Science Teachers & Librarians
- Get permission from your parents/coach to use AI to create a study guide using the rules provided here

Allowed Calculator

This is the acceptable calculators for all Division A events.

Class I - Stand-alone non-graphing, non-programmable, non-scientific 4-function or 5-function calculators. These are the most basic types of calculators. They are limited to the four basic mathematical functions and square roots. They are often cheap, costing as low as a dollar online.



Safety Glasses

Indirect Vent Goggles or Safety glasses are a mandatory safety requirement for some events. It is important to choose the type of protection specific to the task. Competitors are responsible for providing their own protective eyewear. Teams will not be allowed to compete without adequate eye protection.

Category C: Indirect Vent Goggles



Category B: Safety glasses



Compliant Eyewear Categories:

CATEGORY B

- Description: Impact protection. They provide protection from a high inertia particle hazard (high mass or velocity)
- Corresponding ANSI designation/required marking: Z87+.
- Category B eyewear can look either like goggles or safety glasses, but must have the Z87+ marking embossed on them.

CATEGORY C

- Description: Indirect vent chemical/splash protection goggles. These seal tightly to the face completely around the eyes and have indirect vents constructed so that liquids do not have a direct path into the eye (or no vents at all). If you can see through the vent holes from one side to the other, they are NOT indirect vents
- Corresponding ANSI designation/required marking: Z87 (followed by D3 is the most modern designation but, it is not a requirement)

Examples of Non-Compliant Eyewear:

- Face shields/visors are secondary protective devices and are not approved.
- Prescription Glasses containing safety glass should not be confused with safety spectacles. "Safety glass" indicates the glass is made to minimize shattering when it breaks. Unless these glasses bear the Z87+ mark they are not approved for use.

Notes:

1. A goggle that bears the Z87+ mark and is an indirect vent chemical/splash protection goggle will qualify for both Category B & C
2. VisorGogs do not seal completely to the face, but are acceptable as Type C goggles